

Armaan Kanchan

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EDUCATION

Purdue University

B.Eng. in Computer Engineering

Relevant: CompArch, Digital Design, OS, Linux, Computer Networks, Embedded Systems

West Lafayette, IN

May 2026

EXPERIENCE

SanDisk Corporation

Intern

Milpitas, CA

May 2025 – Aug 2025

- Wrote **shell automation scripts (Linux)** for flash DUT connectivity, kernel/IP diagnostics, SPI, USB, and PCIe switch/lane width (Gen 3/4/5) checks, streamlining validation and reducing test time.
- Developed an **OOP GUI in Qt Creator (C++)** integrating these scripts into a single interface, simplifying test execution and accelerating debug for engineers.
- Built an **Ansible** deployment framework to provision/configure **prod/QA/staging** servers (init scripts, DB connections, credentials)—enabling one-command rollouts and cutting deploys from hours to minutes.

Western Digital Corporation

Test Platform Development Intern

San Jose, CA

May 2024 – Aug 2024

- Engineered an **automation test infrastructure** for an internal diagnostic tool, integrating with **Jenkins CI/CD** for continuous verification.
- Implemented **Playwright** UI/system test suites to validate workflows across builds, improving reliability.
- Reduced manual QA by automating end-to-end tests; accelerated release velocity and delivery consistency.

PROJECTS

5-Stage Pipelined CPU (SystemVerilog, FPGA)

Purdue University

Dec 2024 – May 2025

- Designed a **single-cycle** and **5-stage pipelined CPU** on **Intel/Altera Cyclone IV** with hazard detection, forwarding, and cache hierarchy.
- Built MHz-level clocked design with **BPT/BTB** branch prediction to improve pipeline performance.
- Developed a **data cache** and extended to a **multi-core** system with **MSI** coherency; verified via simulation, synthesis, and FPGA prototyping.

AHB-Lite USB 1.0 Verification

Purdue University

Jan 2025 – May 2025

- Designed **USB 1.0 bulk-transfer endpoint** with **AHB-Lite** client interface (modules: USB RX, USB TX, circular FIFO buffer, AHB-Lite satellite).
- Built SystemVerilog testbenches and **QuestaSim** flows; verified host↔endpoint transfers, NRZI encoding at **12 MHz**, bus error handling (**hresp/hready**).

Who Wants to Be a Millionaire (Embedded Systems)

Purdue University

Aug 2024 – Dec 2024

- Implemented a two-player trivia game on **STM32 microcontrollers**, each with a TFT display controlled via **I2C**.
- Enabled inter-board communication using the **USART protocol**, exchanging questions/answers in real time.
- Integrated **GPIO inputs** with a keypad for answer entry and a buzzer to indicate correct/incorrect responses.
- Programmed the entire system in **Embedded C**, handling display updates, synchronization, and scoring logic.

SKILLS

Languages: C/C++, Python, SystemVerilog, Bash

Tools: STM32, ESP32, Qt Creator, Ansible, Jenkins, Playwright, Git, QuestaSim, FPGA (Cyclone IV)

Domains: CPU/Cache/Coherency, PCIe, SPI, USART, I2C, GPIO, USB, Test & Deployment Automation